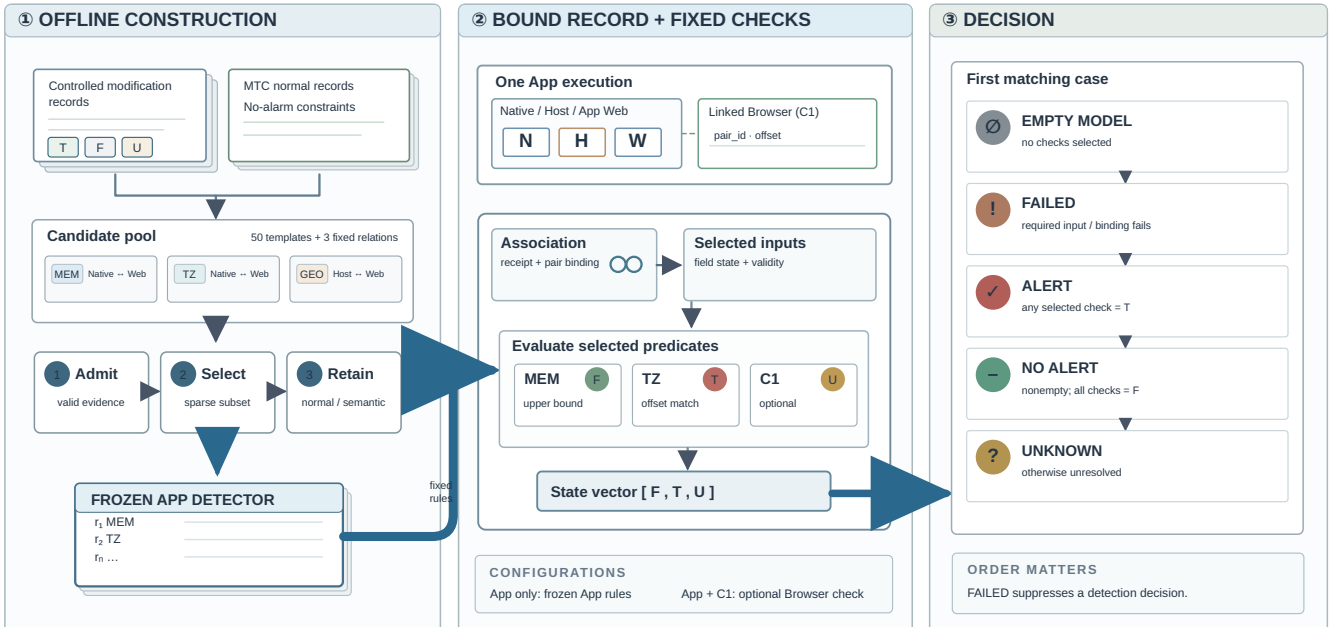
① OFFLINE CONSTRUCTION
Controlled modification records
T F U
MTC normal records
No-alarm constraints
Candidate pool
50 templates + 3 fixed relations
MEM Native ↔ Web
TZ Native ↔ Web
GEO Host ↔ Web
1 Admit
valid evidence
2 Select
sparse subset
3 Retain
normal / semantic
FROZEN APP DETECTOR
r1 MEM
r2 TZ
rn …
fixed rules
② BOUND RECORD + FIXED CHECKS
One App execution
Native / Host / App Web
N H W
Linked Browser (C1)
pair_id · offset
Association
receipt + pair binding
Selected inputs
field state + validity
Evaluate selected predicates
MEM F
upper bound
TZ T
offset match
C1 U
optional
State vector [F , T , U]
CONFIGURATIONS
App only: frozen App rules
App + C1: optional Browser check
③ DECISION
First matching case
EMPTY MODEL
no checks selected
FAILED
required input / binding fails
ALERT
any selected check = T
NO ALERT
nonempty; all checks = F
UNKNOWN
otherwise unresolved
ORDER MATTERS
FAILED suppresses a detection decision.